

D325

Networks

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A network technician at a school district with 14 buildings needs a device that reads MAC addresses and sends each frame only out the port where the destination device is connected. The district just upgraded to fiber uplinks. Which device fits?

- A. A hub
- B. A router
- C. A fiber media converter
- D. A switch

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2. [TRANSFER — reversed stem / different variable]

Hosts on 10.1.1.0/24 need to reach hosts on 10.1.2.0/24. Which device makes this possible?

- A. A hub
- B. A router or Layer 3 switch
- C. A wireless access point
- D. A Layer 2 switch

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

An engineer at a regional bank, which has 40 ATMs, assigns 172.16.40.0/27 to a branch VLAN. The bank's internet circuit is 500 Mbps. How many usable host addresses does the subnet have?

- A. 27
- B. 32
- C. 40
- D. 30

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4. [TRANSFER — reversed stem / different variable]

An administrator must split 192.168.10.0/24 into at least 12 equal subnets. What is the resulting prefix, and how many usable hosts does each subnet have?

- A.** /12 with 12 usable hosts each
- B.** /28 with 14 usable hosts each
- C.** /28 with 16 usable hosts each
- D.** /27 with 30 usable hosts each

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

An audit at a manufacturing company with plants in three states finds that network staff manage switches with Telnet. The switches are two years old and still under support. What should replace Telnet?

- A.** HTTP on port 80
- B.** SNMPv1
- C.** SSH on port 22
- D.** Telnet on a nonstandard port

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6. [TRANSFER — reversed stem / different variable]

A firewall review shows directory queries on TCP 389. Which change encrypts them?

- A.** Move to port 22.
- B.** Move to LDAPS on port 636.
- C.** Move to port 443.
- D.** Keep 389, because directory traffic is always internal.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

A warehouse, which ships 3,000 orders a day, runs a new Cat 6 cable 130 meters from the switch to a packing-station PC. The cable passed a continuity test. The PC shows frequent errors and drops. What is the most likely cause and fix?

- A.** The run exceeds 100 m; add a switch partway or use fiber.
- B.** Upgrade to Cat 6a, which supports 130 m runs.
- C.** Shorten the run to 130 feet.
- D.** The cable is fine because it passed continuity.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

A clinic with 35 staff puts front-desk PCs in VLAN 10 and billing PCs in VLAN 20 on the same switch. The switch has 48 ports. Front-desk staff now cannot reach the billing server. What is required for them to communicate?

- A.** A router or Layer 3 switch to route between the VLANs
- B.** Nothing; VLANs on the same switch always talk freely.
- C.** A switch with more than 48 ports
- D.** A hub between the two VLANs

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

A retail headquarters with 400 staff has two routers, 10.0.0.1 and 10.0.0.2. All PCs use 10.0.0.1 as their default gateway. When that router failed last week, every PC lost internet access even though the second router was fine. The routers were bought from different vendors. What fixes this?

- A.** Configure a first-hop redundancy protocol such as VRRP with a shared virtual gateway IP.
- B.** Replace both with routers from the same vendor.
- C.** Add a third router with its own IP.
- D.** Point half the PCs to each router.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

An office building's three 2.4 GHz access points, installed by a contractor last spring, are set to channels 1, 3, and 5. Users near the middle of the floor see slow speeds. What channel plan should be used?

- A.** 1, 3, and 5 at higher power
- B.** 2, 4, and 6
- C.** 1, 6, and 11
- D.** All three on channel 6

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

A nonprofit, whose website was redesigned in March, moves its email to a cloud provider. Incoming mail keeps going to the old server. The web server's A record is correct. Which DNS record must be updated?

- A.** A
- B.** PTR
- C.** MX
- D.** CNAME

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	____	

12. [TRANSFER — reversed stem / different variable]

A security tool must find the hostname for the IP address 203.0.113.25. Which DNS record type answers this?

- A.** A
- B.** AAAA
- C.** MX
- D.** PTR

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	____	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer D Concept 1

A network technician at a school district ~~with 14 buildings~~ needs a device that reads MAC addresses and sends each frame only out the port where the destination device is connected. ~~The district just upgraded to fiber uplinks.~~ Which device fits?

Target word: device that forwards by MAC **Decides:** MAC-based forwarding is Layer 2.

A	Adjacent category	A hub repeats every signal out all ports.
B	Confusion set	Routers forward by IP address at Layer 3.
C	Camouflage	Changes the medium; it does not make forwarding decisions.
D	Correct	Switches learn MAC addresses and forward frames at Layer 2.

2. Answer B Concept 1 • transfer

Hosts on 10.1.1.0/24 need to reach hosts on 10.1.2.0/24. Which device makes this possible?

Target word: device that connects different networks **Decides:** Moving between subnets requires Layer 3 forwarding.

A	Adjacent category	Layer 1; no forwarding decisions.
B	Correct	Forwards packets between networks by IP address.
C	Confusion set	Bridges wireless clients into one network.
D	Inverted operation	Forwards within one network only.

3. Answer D Concept 2

An engineer at a regional bank, ~~which has 40 ATMs,~~ assigns 172.16.40.0/27 to a branch VLAN. ~~The bank's internet circuit is 500 Mbps.~~ How many usable host addresses does the subnet have?

Target word: usable hosts in a /27 **Decides:** $32 - 27 = 5$ host bits.

A	Wrong stem number	The prefix length.
B	Dropped element	Leaves out network and broadcast addresses.
C	Camouflage	The ATM count does not decide the subnet size.
D	Correct	$2^5 - 2 = 30$.

4. Answer B Concept 2 • transfer

An administrator must split 192.168.10.0/24 into at least 12 equal subnets. What is the resulting prefix, and how many usable hosts does each subnet have?

Target word: solve for prefix from subnets needed **Decides:** $2^4 = 16 \geq 12$, so borrow 4 bits.

A	Wrong stem number	Uses the subnet count as the prefix.
B	Correct	Borrowing 4 bits gives 16 subnets; 4 host bits leave $2^4 - 2 = 14$.
C	Dropped element	Forgets to subtract the 2 reserved addresses.
D	Inverted operation	Borrowing 3 bits gives only 8 subnets.

5. Answer C Concept 3

An audit at a manufacturing company with plants in three states finds that network staff manage switches with Telnet. ~~The switches are two years old and still under support.~~ What should replace Telnet?

Target word: secure management protocol **Decides:** Telnet sends credentials in clear text.

A	Adjacent category	Also clear text.
B	Confusion set	Uses clear-text community strings; SNMPv3 is the secure version.
C	Correct	Encrypts the management session.
D	Dropped element	Moving the port keeps the traffic unencrypted.

6. Answer B Concept 3 • transfer

A firewall review shows directory queries on TCP 389. Which change encrypts them?

Target word: secure version of LDAP **Decides:** LDAP 389 is clear text; LDAPS uses 636.

A	Adjacent category	SSH, a different protocol.
B	Correct	LDAP over TLS encrypts directory traffic.
C	Confusion set	443 is HTTPS, not directory access.
D	Absolute language	Internal traffic can still be captured.

7. Answer A Concept 4

A warehouse, ~~which ships 3,000 orders a day,~~ runs a new Cat 6 cable 130 meters from the switch to a packing-station PC. ~~The cable passed a continuity test.~~ The PC shows frequent errors and drops. What is the most likely cause and fix?

Target word: cause of errors on a long copper run **Decides:** 130 m exceeds the 100 m copper limit.

A	Correct	Attenuation beyond 100 m causes errors.
B	Adjacent category	Cat 6a is still limited to 100 m.
C	Wrong stem number	Mixes units; the fix is about the 100 m limit.
D	Camouflage	Continuity does not test for length-related signal loss.

8. Answer A Concept 5

A clinic ~~with 35 staff~~ puts front-desk PCs in VLAN 10 and billing PCs in VLAN 20 on the same switch. ~~The switch has 48 ports.~~ Front-desk staff now cannot reach the billing server. What is required for them to communicate?

Target word: inter-VLAN communication **Decides:** Different VLANs are separate networks.

A	Correct	Traffic between VLANs must be routed.
B	Absolute language	VLANs are separate broadcast domains.
C	Camouflage	Port count does not affect VLAN separation.
D	Adjacent category	Layer 1 devices do not route.

9. Answer A Concept 6

A retail headquarters with 400 staff has two routers, 10.0.0.1 and 10.0.0.2. All PCs use 10.0.0.1 as their default gateway. When that router failed last week, every PC lost internet access even though the second router was fine. ~~The routers were bought from different vendors.~~ What fixes this?

Target word: gateway failover **Decides:** Hosts point to one physical gateway.

A	Correct	Hosts use the virtual IP, which moves to the surviving router.
B	Camouflage	VRRP is vendor-neutral; the brand is not the issue.
C	Confusion set	Another IP does not give hosts automatic failover.
D	Dropped element	Half the PCs still fail when one router dies.

10. Answer C Concept 7

An office building's three 2.4 GHz access points, ~~installed by a contractor last spring,~~ are set to channels 1, 3, and 5. Users near the middle of the floor see slow speeds. What channel plan should be used?

Target word: non-overlapping channels **Decides:** Only 1, 6, and 11 do not overlap.

A	Inverted operation	More power increases overlap.
B	Adjacent category	Still overlapping.
C	Correct	The three non-overlapping 2.4 GHz channels.
D	Confusion set	Same channel means they share airtime.

11. Answer C Concept 8

A nonprofit, ~~whose website was redesigned in March,~~ moves its email to a cloud provider. Incoming mail keeps going to the old server. ~~The web server's A record is correct.~~ Which DNS record must be updated?

Target word: record for mail routing **Decides:** Mail delivery uses MX records.

A	Camouflage	The A record for the website is not what routes mail.
B	Confusion set	Used for reverse lookups.
C	Correct	MX records tell other servers where to deliver mail.
D	Adjacent category	Creates an alias; it does not route mail.

12. Answer D Concept 8 • transfer

A security tool must find the hostname for the IP address 203.0.113.25. Which DNS record type answers this?

Target word: record for reverse lookup **Decides:** IP to name is a reverse lookup.

A	Inverted operation	Maps a name to an address, the opposite direction.
B	Adjacent category	Maps a name to an IPv6 address.
C	Confusion set	Names mail servers.
D	Correct	PTR records map an address back to a name.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	4	
Wrong stem number	A figure copied from the prompt that is not the answer.	3	
Adjacent category	The right idea in the wrong bucket.	9	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	4	
Absolute language	Always, never, guarantee — where judgment applies.	2	
Confusion set	Every option is real; the test is exact role-matching.	8	
Camouflage	Answers a stem fact that was never the question.	6	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	OSI layers of network devices	Hubs repeat signals at Layer 1; switches forward frames by MAC address at Layer 2; routers forward packets by IP address between networks at Layer 3.	Saying a standard switch routes between subnets, or a router forwards by MAC.	What address a device reads to make its forwarding decision.
2	Subnetting	Usable hosts = $2^{(\text{host bits})} - 2$; subnets created = $2^{(\text{borrowed bits})}$.	Forgetting the network and broadcast addresses, or mixing up host bits and prefix bits.	A CIDR prefix, a host count, or a number of subnets needed.
3	Secure protocol replacements	Use SSH (22) instead of Telnet (23), SFTP or FTPS instead of FTP, HTTPS (443) instead of HTTP, LDAPS (636) instead of LDAP (389), and SNMPv3 instead of v1 or v2c.	Keeping the insecure protocol because it works.	Credentials or data sent in clear text.
4	Twisted-pair distance limit	Copper Ethernet (Cat 5e, 6, 6a) runs are limited to 100 meters; longer runs need a switch in between or fiber.	Blaming the cable category when the run is simply too long.	A cable length over 100 m and intermittent errors.
5	VLANs and trunking	VLANs split one switch into separate broadcast domains; links carrying several VLANs between switches are trunks using 802.1Q tags; traffic between VLANs needs a router or Layer 3 switch.	Expecting devices in different VLANs to talk without routing.	Two VLANs that cannot reach each other, or VLAN traffic across switches.
6	Default gateway redundancy	First-hop redundancy protocols (VRRP, HSRP) let two routers share a virtual gateway IP so hosts keep working if one router fails.	Assuming a second router with a different IP gives automatic failover to hosts.	Hosts lose connectivity when one router fails despite a backup router.
7	2.4 GHz channel planning	In the 2.4 GHz band, only channels 1, 6, and 11 do not overlap (in North America); neighboring APs should use different ones of these.	Placing APs on adjacent channels such as 1, 2, and 3.	Nearby access points with interference complaints.
8	DNS record types	A maps a name to IPv4, AAAA to IPv6, CNAME is an alias to another name, MX names mail servers, PTR maps an IP back to a name, TXT holds text such as SPF.	Using a CNAME or A record to route email.	Mail delivery, reverse lookup, or an alias.

Course scope (public description): Networks introduces skills in configuring networking components and a network infrastructure. Learners will gain skills in optimizing network operations for availability, performance, and security, and in troubleshooting network issues. The course prepares learners for the CompTIA Network+ certification exam. Network and Security - Foundations is a prerequisite for this course.